Derivative

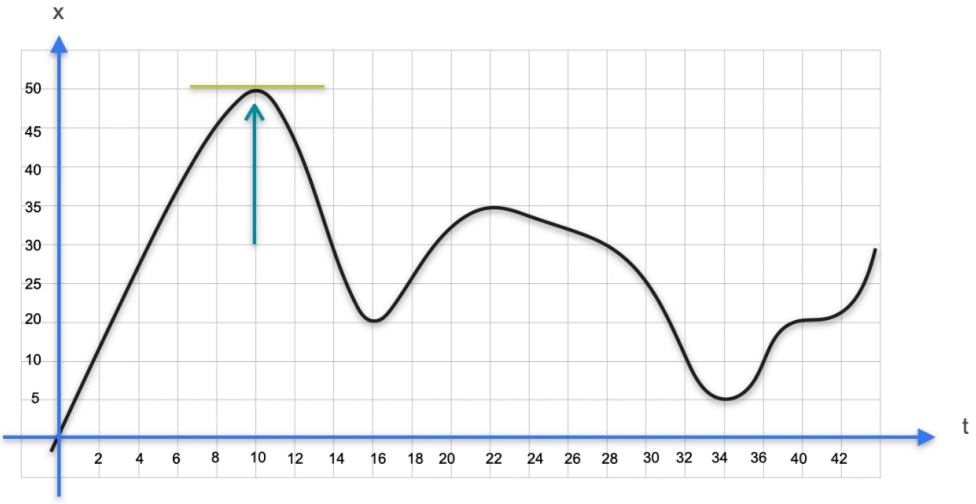
Derivatives are important in Machine Learning because they are used to optimize functions, particularly to minimize or maximize their values.

Velocity is a creative analogy to think of the derivative

- Derivative is a continuous rate of change of a function.

Minima and Maxima

If we want to find the minimum and maximum points in the function, they occur at one of the points where the derivative is 0 or where the tangent line is horizontal.



Notation

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{\Delta y}{\Delta x}$$

$$\text{Slope at a point} = \frac{dy}{dx}$$

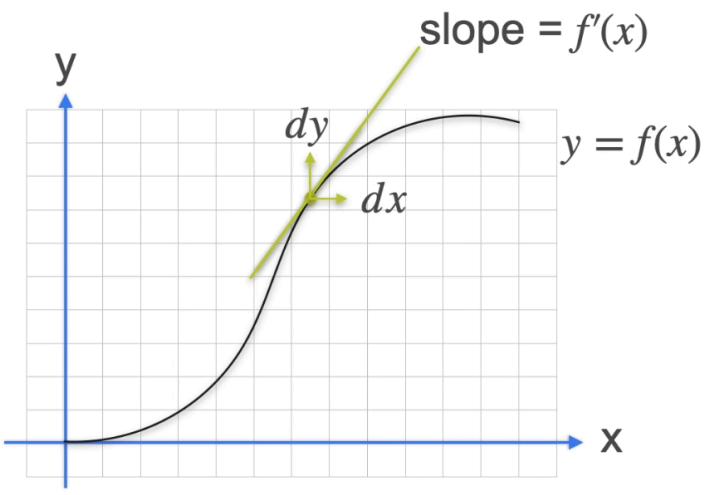
Lagrange's and Leibniz's Notation

$$y = f(x)$$

Derivative of f is expressed as:

$f'(x)$ **Lagrange's notation**

$\frac{dy}{dx} = \frac{d}{dx} f(x)$ **Leibniz's notation**



Inverse

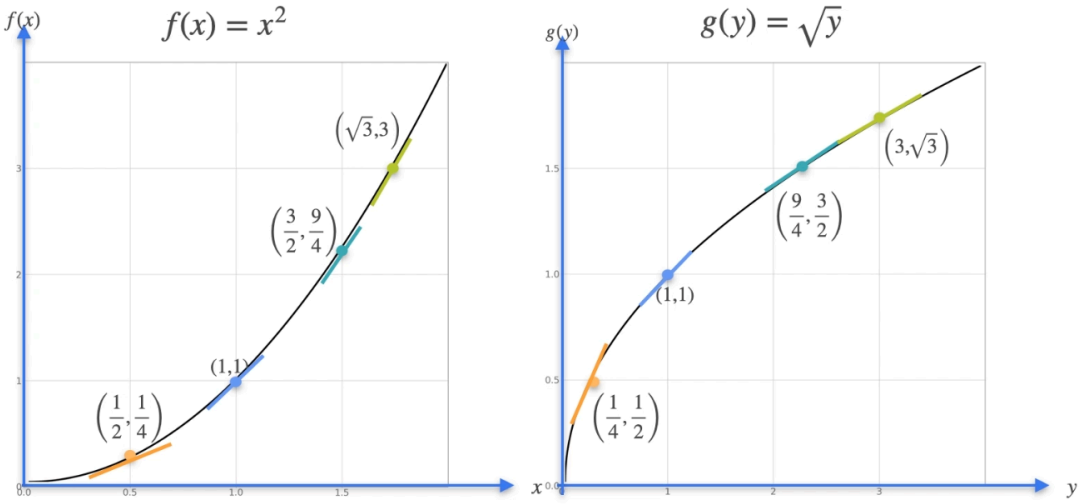
It undoes what F does

x and x are inverses

$$x = x$$

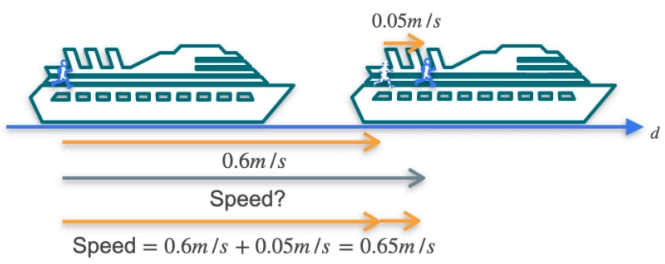
$$x = x$$

Derivative of the inverse



$$x = \overline{x}$$

Sum rule



- Speed of the boat: 0.6 m/s
- Speed of the child inside the boat: 0.05 m/s
- Both travel in the same direction.

What is the speed of the child with respect to the earth?

Answer: 0.65 m/s

Velocity can be added if they have the same directions

The product rule

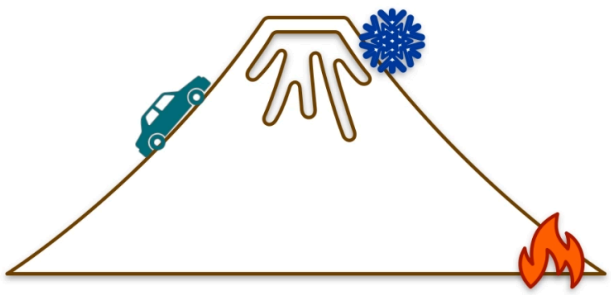
$$=$$

$$=$$

The chain rule

$$\frac{d}{d} = \frac{d}{d}\,\frac{d}{d}\,\frac{d}{d}$$

Idea of the chain rule



Temperature changes w.r.t. height $\frac{dT}{dh}$

height changes w.r.t. time $\frac{dh}{dt}$

Temperature changes w.r.t. time $\frac{dT}{dt}$

$$\frac{dT}{dt} = \frac{dT}{dh}\,\frac{dh}{dt}$$